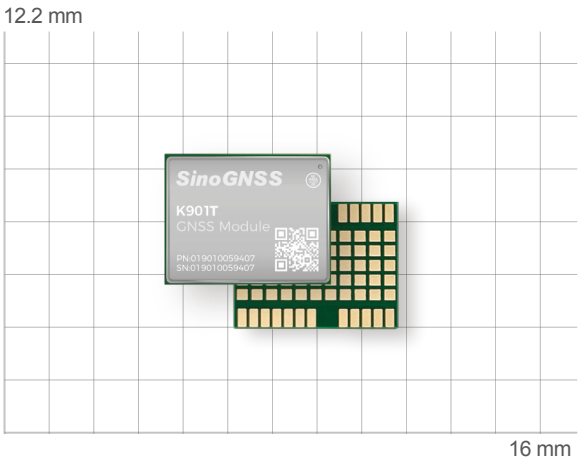
K901T GNSS Module

K Series GNSS Module | Ver.2025.06.17



Size(W × L × H)**:** 12.2 mm × 16 mm × 2.5 mm
Weight: 1.0 g

K901T
GNSS Module

Next-Gen QC7820 SoC Technology

The K901T incorporates ComNav's QC7820 SoC, an advanced multi-frequency GNSS processor designed for reliable signal tracking across all operational navigation constellations. The chipset's innovative architecture ensures robust performance in diverse signal environments.

Features

- Support GPS, BDS-2, BDS-3, GLONASS, GALILEO, NAVIC, QZSS and SBAS
- RF&BB Integrated High-Precision GNSS SoC Chip
- Brand New Dual Core CPU Structure
- 5ns Timing Accuracy
- Support PPP-B2b, PPP-HAS Service

High Precision Timing Technology

The K901T GNSS module provides nanosecond-level timing precision for mission-critical infrastructure. With advanced multi-band technology, it achieves <5ns timing accuracy in open-sky conditions without relying on external correction services.

Multi Anti-Interference Technologies

The K901T module features an advanced internal adaptive anti-interference system incorporating wideband reception, narrowband suppression, and continuous-wave rejection technologies. Its intelligent algorithm effectively mitigates all types of RF interference, ensuring high-quality observation data even in the most challenging electromagnetic environments.

Application



Easy Integration

With its space-saving SMT package and ultra-low power operation, the K901T maintains full pin compatibility with industry-standard GNSS modules, simplifying design-in and reducing development time.

SinoGNSS

©2025, ComNav Technology Ltd. All rights reserved. SinoGNSS is the official trade mark of ComNav Technology Ltd., registered in People's Republic of China, EU, USA and Canada. All other trademarks are the property of their respective owners. (May,2024).

K901T GNSS Module

K Series GNSS Module | Ver.2025.06.17

Signal Tracking

GPS	L1C/A, L2P, L2C, L5, L1C
BDS-2	B1I, B2I, B3I
BDS-3	B1I, B3I, B1C, B2a, B2b
GALILEO	E1, E5b, E5a, E5 AltBoC*, E6c*
GLONASS	G1, G2, G3*
SBAS	L1C/A, L5
QZSS	L1C/A, L2C, L5, L1C
NAVIC*	L5

Performance Specifications

Cold Start	< 20 s (Adding Acceleration Capture Module)
Hot Start (with RTC)	< 10 s (Typical)
Signal Reacquisition	< 1 s
Initialization Reliability	> 99.9%
Velocity Accuracy	≤ 0.02 m/s
Time Accuracy	5 ns
PPP Convergence Time	<15 min[1]

Positioning Specifications

Post Processing	2.5 mm + 1 ppm Horizontal 5 mm + 1 ppm Vertical
DGPS	< 0.4 m RMS
SBAS	1 m 3D RMS
Standalone	1.5 m Horizontal; 3 m Vertical
PPP	0.1 m Horizontal; 0.2 m Vertical

Communications Interfaces

UART	x 3
PPS	x 1
EVENT	x 1
I2C	x 1

Data Format

Correction data I/O	RTCM 2.X, 3.X
Position data output	- ASCII (NMEA-0183): GGA, GSA, GSV, RMC, HDT, VHD, ZDA, VTG, GST, GLL -Binary: ComNav Binary -Position Output Rate: 1 Hz, 2 Hz, 5 Hz,10 Hz, 20 Hz*, 50 Hz*

Antenna Interface

Impedance Match	50 Ω
LNA Power (External)	+ 3.3V ~ + 5V ± 5%VDC @ 0-100 mA
LNA Gain	20 ~ 35 dB (Suggested)

Physical

Size (W × L × H)	12.2 mm × 16 mm × 2.5 mm
Hardware Interface	LGA 24 pin
Weight	1.0 g

Environmental

Working Temperature	-40 °C to + 85 °C
Storage Temperature	-55 °C to + 125 °C

Electrical

Input Voltage	+ 3.3 V ± 5% DC
Power Consumption	0.35 W

Software Tools

ComNav Compass Receiver Utility

Compass Solution Software

Note: Items marked with *are only support by specific firmware.

[1] The laboratory test results may be affected by experimental environment and are provided for reference purposes.

SinoGNSS

ComNav Technology Ltd.

Building 2, No. 618 Chengliu Middle Road, 201801 Shanghai, China

Web: www.comnavtech.com

Email: sales@comnavtech.com

Tel : +86 21 64056796 Fax: +86 21 54309582